

JIANGHUI SUN

AI SYSTEMS / AGENT MEMORY / MULTIMODAL LEARNING
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EDUCATION

University of Illinois Urbana-Champaign	Aug 2026 - Expected Dec 2027
Master of Computer Science (MCS)	
Xi'an Jiaotong-Liverpool University	Sep 2022 - Jul 2026
B.Sc. in Information and Computing Science GPA: 3.8/4.0	

RESEARCH & INDUSTRY EXPERIENCE

Research Assistant UIUC SSAIL	Current
Assisting with system optimizations for prefill/decode (P/D)-disaggregated LLM inference for multi-turn agentic tasks.	
Research Assistant Xi'an Jiaotong-Liverpool University	Nov 2025 - Jan 2026
Fine-grained Emoji Understanding	
Built a DINOv2-Qwen-VL framework for emoji understanding; designed adapter-based visual alignment and connected features to the language decoder to describe subtle facial expressions.	
Research Assistant Xi'an Jiaotong-Liverpool University	Sep 2025 - Nov 2025
EmoSound: Emotion-Aware Audio Accompaniment	
Developed a multimodal agent for visually impaired users, combining emotion-aligned retrieval and audio generation; fine-tuned CLIP with lightweight cross-attention soft prompts for low-resource emotion recognition.	
ML Algorithm Engineering Intern RockControl Technology Group	Jun 2024 - Aug 2024
Modeled 40k+ records of air quality, emissions, and weather using RNN, LSTM, GRU, and Transformers in PyTorch for 6-, 12-, and 24-hour forecasts of PM2.5, O3, and AQI.	
Reduced RMSE by approximately 12% with LSTM versus baseline RNN, achieving the most stable performance across forecast horizons.	

SELECTED PROJECTS

Knowledge Organization for Agent Memory Evolution	In progress
CS546 Team Research Project UIUC	
Investigating whether organizing identical knowledge improves agents' learning of selective memory revision, comparing flat and semantically organized memory under matched content and feedback budgets.	
Planning reinforcement-learning evaluation of learning gains, stale-memory correction, and valid-knowledge preservation; evaluation plan includes STALE, AgentStream, and MemoryAgentBench.	
Cross-modal Audio Generation for Visually Impaired Users	Jun 2025 - Sep 2025
Built an image-to-audio web application with training-free multimodal RAG , a cross-modal knowledge graph, dual-vector databases, and Qwen-based aggregation of image tags and descriptions.	
Daily Reading Tracker	Mar 2025 - Jun 2025
Implemented the reading-log module across the front end and back end of a deployed full-stack Java application; contributed technical documentation.	

PUBLICATION

Sun, J., et al. "EmoSound: A Multimodal AI Agent Framework for Emotion-Aware Audio Accompaniment of Emoticons." *Advances in Brain-Inspired Cognitive Systems*, LNCS 16568, Springer, 2026. **Accepted; forthcoming**. Presented at BICS 2025.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, SQL
AI/ML: PyTorch, Transformers, scikit-learn, Computer Vision, NLP, LoRA/PEFT, LLMs, RAG, LangChain
Backend & Tools: FastAPI, Spring Boot, REST APIs, Docker, Git/GitHub, Jupyter, Postman, n8n